

NMAP PORT SCANNER USING PYTHON

ABSTRACT

Network security is an important aspect of modern computing systems. Identifying open ports and active services helps administrators monitor and secure their networks. This project implements a simple Port Scanner using Python and the Nmap library. The application provides a user-friendly command-line interface that allows users to perform SYN ACK scans, UDP scans, and Comprehensive scans on a specified IP address. The scanner displays information such as host status, available protocols, and open ports. This project demonstrates the integration of Python programming with network security tools and provides a practical introduction to cybersecurity concepts.

INTRODUCTION

Port scanning is a technique used to discover open ports and services running on a network host. Open ports can reveal information about active applications and network services. Nmap (Network Mapper) is one of the most widely used network scanning tools in cybersecurity.

This project uses Python and the python-nmap library to automate Nmap scans. Users can enter an IP address and select the desired scan type through a menu-driven interface.

OBJECTIVES

1. To understand the fundamentals of network scanning.
2. To automate Nmap operations using Python.
3. To identify active hosts and open ports.
4. To provide a simple interface for different scan types.
5. To gain practical knowledge of cybersecurity and ethical hacking tools.

SOFTWARE REQUIREMENTS

- Operating System: Kali Linux / Linux
- Programming Language: Python 3
- IDE: Visual Studio Code
- Tool: Nmap
- Library: python-nmap

HARDWARE REQUIREMENTS

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB
- Storage: 10 GB free space
- Network Connectivity

METHODOLOGY

Step 1: User Input

The program accepts an IP address from the user.

Step 2: Scan Selection

The user chooses one of the following scan types:

- SYN ACK Scan
- UDP Scan
- Comprehensive Scan

Step 3: Scan Execution

The selected Nmap command is executed using the Python Nmap library.

Step 4: Result Processing

The scan results are collected and analyzed.

Step 5: Output Display

The application displays:

- Nmap version
- Host status
- Available protocols
- Open ports

SCAN TYPES IMPLEMENTED

1. SYN ACK Scan

Command Used:

`nmap -sS`

Purpose:

- Detects open TCP ports.
- Fast and efficient scanning method.
- Commonly used during reconnaissance.

2. UDP Scan

Command Used:

```
nmap -sU
```

Purpose:

- Identifies open UDP ports.
- Useful for discovering services such as DNS and DHCP.

3. Comprehensive Scan

Command Used:

```
nmap -sS -sV -sC -A -O
```

Purpose:

- Service version detection.
- Operating system detection.
- Default script execution.
- Detailed host information gathering.

ALGORITHM

1. Start the program.
2. Initialize the Nmap PortScanner object.
3. Accept IP address from the user.
4. Display scan menu.
5. Read user choice.
6. Execute corresponding scan.
7. Collect scan results.
8. Display host status and open ports.
9. End program.

PROGRAM OUTPUT

Example:

Welcome, this is a simple nmap automation tool

Please enter the IP address: 172.30.251.8

Select scan type: 1. SYN ACK Scan 2. UDP Scan 3. Comprehensive Scan

IP Status: Up Protocols: [] No TCP ports found.

ADVANTAGES

- Easy to use.
- Automates Nmap scanning.
- Useful for learning cybersecurity.
- Supports multiple scan types.
- Provides quick network information.

LIMITATIONS

- Command-line interface only.
- Supports a single host at a time.
- Limited error handling.
- Depends on Nmap installation.
- Requires appropriate permissions for advanced scans.

FUTURE ENHANCEMENTS

- Graphical User Interface (GUI) using Tkinter.
- Export reports to PDF and CSV.
- Multi-host scanning support.
- Real-time network monitoring.
- Vulnerability assessment integration.
- Scan history management.

APPLICATIONS

- Network Administration
- Security Auditing
- Penetration Testing
- Cybersecurity Education
- Troubleshooting Network Services

CONCLUSION

The Nmap Port Scanner using Python successfully automates network scanning tasks through a simple command-line interface. The project demonstrates how Python can be integrated with Nmap to perform SYN, UDP, and comprehensive scans. It provides valuable experience in network security, cybersecurity fundamentals, and Python programming. The project serves as an effective learning tool for students interested in ethical hacking and network administration.